GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

Qualification Pack

Supervisor - 2 and 3 Wheeler Electric Vehicle Services

QP Code: MSU/ASC/Q0101

Version: 1.0

NSQF Level: 5

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MSU/ASC/Q0101: Supervisor - 2 and 3 Wheeler Electric Vehicle Services

Brief Job Description

On completion of this course, candidates will be skilled in diagnosing and repairing mechanical, electrical, and electronic vehicle components, including battery management for 2- and 3-wheeler electric vehicles. They will be able to manage service workshops, lead repair teams, and handle customer requests and complaints efficiently. This qualification equips individuals with both the technical expertise and managerial skills necessary to excel in vehicle maintenance and customer service.

Personal Attributes

A candidate for this role should possess strong problem-solving skills and a sharp mind for diagnosing complex vehicle issues. They should be technically adept, with excellent attention to detail in both mechanical and electrical repairs. Leadership and organizational skills are essential for managing teams and workshops effectively. Additionally, strong communication and customer service abilities are required to handle customer requests and complaints professionally. Adaptability, time management, and a proactive attitude are key traits for staying efficient in this fast-paced, evolving field.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. MSU/ASC/N0101: Perform fault diagnosis on vehicles
2. MSU/ASC/N0102: Carry out service, repair and overhauling of mechanical aggregates in vehicles
3. MSU/ASC/N0103: Carry out service, repair and overhauling of electrical and electronic aggregates in vehicles
4. MSU/ASC/N0104: Carry out battery swapping and management for 2 and 3 wheeler Electric vehicle
5. MSU/ASC/N0105: Manage vehicle service and repair workshops
6. MSU/ASC/N0107: Manage the vehicle service and repair teams
7. MSU/ASC/N0106: Deal with Customer's Request and Complaints
8. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Automotive

Qualification Pack

Sub-Sector	Electric Vehicle
Occupation	Automotive Servicing and Repair
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3115.0602, 3322.2501
Minimum Educational Qualification & Experience	10th grade pass with 4 Years of experience relevant experience OR 12th grade Pass with 2 Years of experience relevant experience OR 12th grade pass with 1 year NTC/ NAC with 1 Year of experience relevant experience OR UG in any field (completed 2nd year of under graduation after 12th in the field of Automotive/EV) OR Pursuing 2nd year of UG (3-year/ 4-years UG and continuing education in the field of Automotive/EV)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-05-AU-00773-2023-V1-MSU
NQR Version	1

MSU/ASC/N0101: Perform fault diagnosis on vehicles

Description

This OS unit is about managing the fault diagnosis process. It involves gathering fault information, preparing for diagnostic tests, diagnosing the issue, and concluding with a suitable repair solution. Each step ensures a systematic approach to identifying and resolving system faults effectively.

Scope

The scope covers the following :

- Gather the information about the fault
- Prepare to perform diagnostic tests
- Diagnose the fault
- Conclude the repair solution for the fault

Elements and Performance Criteria

Gather the information about the fault

To be competent, the user/individual on the job must be able to:

PC1. Gather information about the fault from the customer through a probing technique

PC2. Check the vehicle history available on Enterprise Resource Planning (ERP)

PC3. Assess the ambient condition of the vehicle during the fault

PC4. Fill the job card with all necessary information about the fault, ambient condition and service history

Prepare to perform diagnostic tests

To be competent, the user/individual on the job must be able to:

PC5. Adhere to the safety aspects while working on a vehicle. Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard, etc

PC6. Check the functioning of vehicle systems and note abnormal behaviour or malfunctioning

PC7. Conduct the visual vehicle inspection to identify whether any external impact, bend, leak, incorrect level or wear and tear are the reasons for the faults

PC8. Check vehicle performance by test-driving the vehicle

PC9. Validate the fault by simulating or reproducing it under the noted ambient condition

PC10. Determine symptoms and precise location of the fault(s) in vehicle systems

PC11. Collect evidence such as photos, audio or videos in accordance with organizational requirements

PC12. Prepare the vehicle for diagnostic test

PC13. Organise for relevant workshop tools, measuring devices and equipment required for fault diagnosis

PC14. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment

Diagnose the fault

To be competent, the user/individual on the job must be able to:

PC15. Take precautions to avoid damage to the vehicle and its components while working on the vehicle

PC16. Check if the fault can be rectified without the replacement of any components as per the Original Equipment Manufacturer’s (OEM) Standard Operating Procedures (SOPs)Faults: e.g., improper charging, loose/poor contacts of pins in wiring harness connectors and their connection, improper driving style etc

PC17. Use the appropriate device and equipment to inspect and diagnose various direct and indirect deficiencies/faults in various systems, Various systems: e.g., body management systems, braking, stability control systems, drive line systems, battery management system, telematics systems, etc

PC18. Identify and isolate the fault and/or defective parts by performing tests using measuring devices, testers, diagnostic tools, software etc. as per the SOP provided by OEM

PC19. Perform diagnostic tasks on the High Voltage (HV) system in various states.States: without isolation of the HV systems (for general and mechanical tasks), in the non-live state of the HV systems (for electrical work), in the live state of the HV system (for troubleshooting and replacing parts)

PC20. Determine the possible repair solution based on own observations, test results, vehicle/component specifications checklists, service/repair/diagnostic manual, technical information, etc. provided by OEM

PC21. Inform the concerned person(s) of any failure or malfunctions, in the vehicle, where the solution is not available, along with preliminary diagnostic details

Conclude the repair solution for the fault

To be competent, the user/individual on the job must be able to:

PC22. Maintain the documentation related to inspections and diagnoses performed on the vehicle

PC23. Interpret the inspection, measurement, and test results as required

PC24. Compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements

PC25. Find possible options for repair or replacement

PC26. Prepare the final proposals regarding repair or replacement requirements with justification

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Standard Operating Procedures (SOPs) for receiving two/three wheeler electric vehicles, opening a job card, checking service history, invoicing, vehicle delivery, handling complaints, etc

KU2. Different components and aggregates of 2 or 3 wheeler electric vehicles as well as auto component manufacturer's specifications for the same

KU3. Basic technology used in various systems and components of the two/three wheeler electric vehicle system and their functioning

KU4. Systems and Components: e.g. brakes, suspension, steering, hub drive/chain drive and electrical machines and devices used in electric vehicles such as generators, DC/AC and DC/DC converters, drive motor/hub, charging systems, regenerative braking, etc.

KU5. Interconnection of systems with each other and the effect of one system on other systems

KU6. Common faults that occur in 2 and 3 wheeler electric vehicles and their symptoms and causes

KU7. Various sources of information available for diagnostic tests, visual inspections, test drives, vehicle/equipment manufacturer specifications, and tolerance limits of components

KU8. Standard Operating Procedures (SOP) and the checklists recommended by the Original Equipment Manufacturer (OEM) /auto component manufacturer/dealer for diagnosing the fault

KU9. Solutions typically applied to common faults in mechanical, electrical and electronic systems of the vehicle

KU10. Safety aspects recommended by OEM and dealer while working on a diagnosis, troubleshooting and root cause analysis on various aggregates

KU11. Five safety rules for electrical work on HV systems before starting the work i.e., isolate, safeguard reconnection, verify the non-live state, earth or short-circuit and shroud or safeguard adjacent live parts

KU12. Procedure to work on the High Voltage (HV) systems that do not require isolation, troubleshooting and replacing parts on the active HV system

KU13. Tools/equipment for use for testing and diagnosis in line with OEM specifications, such as special service tools, measuring instruments, volt meters, ammeters, ohmmeters, battery testers, dedicated and computer-based diagnostic equipment, oscilloscopes etc

KU14. Common errors/defects in tools/equipment used for testing and diagnosis and the way to handle these errors or defects

KU15. Optimised repair solution for the diagnosed fault of the vehicle

KU16. Importance of proposal and proposing to conclude repair solution

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Gather the information about the fault	**5**	**10**	**-**	**-**
PC1. Gather information about the fault from the customer through a probing technique	-	-	-	-
PC2. Check the vehicle history available on Enterprise Resource Planning (ERP)	-	-	-	-
PC3. Assess the ambient condition of the vehicle during the fault	-	-	-	-
PC4. Fill the job card with all necessary information about the fault, ambient condition and service history	-	-	-	-
Prepare to perform diagnostic tests	**5**	**10**	**-**	**-**
PC5. Adhere to the safety aspects while working on a vehicle. Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard, etc	-	-	-	-
PC6. Check the functioning of vehicle systems and note abnormal behaviour or malfunctioning	-	-	-	-
PC7. Conduct the visual vehicle inspection to identify whether any external impact, bend, leak, incorrect level or wear and tear are the reasons for the faults	-	-	-	-
PC8. Check vehicle performance by test-driving the vehicle	-	-	-	-
PC9. Validate the fault by simulating or reproducing it under the noted ambient condition	-	-	-	-
PC10. Determine symptoms and precise location of the fault(s) in vehicle systems	-	-	-	-
PC11. Collect evidence such as photos, audio or videos in accordance with organizational requirements	-	-	-	-
PC12. Prepare the vehicle for diagnostic test	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Organise for relevant workshop tools, measuring devices and equipment required for fault diagnosis	-	-	-	-
PC14. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment	-	-	-	-
Diagnose the fault	**10**	**25**	**-**	**-**
PC15. Take precautions to avoid damage to the vehicle and its components while working on the vehicle	-	-	-	-
PC16. Check if the fault can be rectified without the replacement of any components as per the Original Equipment Manufacturer’s (OEM) Standard Operating Procedures (SOPs)Faults: e.g., improper charging, loose/poor contacts of pins in wiring harness connectors and their connection, improper driving style etc	-	-	-	-
PC17. Use the appropriate device and equipment to inspect and diagnose various direct and indirect deficiencies/faults in various systems, Various systems: e.g., body management systems, braking, stability control systems, drive line systems, battery management system, telematics systems, etc	-	-	-	-
PC18. Identify and isolate the fault and/or defective parts by performing tests using measuring devices, testers, diagnostic tools, software etc. as per the SOP provided by OEM	-	-	-	-
PC19. Perform diagnostic tasks on the High Voltage (HV) system in various states.States: without isolation of the HV systems (for general and mechanical tasks), in the non-live state of the HV systems (for electrical work), in the live state of the HV system (for troubleshooting and replacing parts)	-	-	-	-
PC20. Determine the possible repair solution based on own observations, test results, vehicle/component specifications checklists, service/repair/diagnostic manual, technical information, etc. provided by OEM	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Inform the concerned person(s) of any failure or malfunctions, in the vehicle, where the solution is not available, along with preliminary diagnostic details	-	-	-	-
Conclude the repair solution for the fault	**10**	**25**	**-**	**-**
PC22. Maintain the documentation related to inspections and diagnoses performed on the vehicle	-	-	-	-
PC23. Interpret the inspection, measurement, and test results as required	-	-	-	-
PC24. Compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements	-	-	-	-
PC25. Find possible options for repair or replacement	-	-	-	-
PC26. Prepare the final proposals regarding repair or replacement requirements with justification	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0101
NOS Name	Perform fault diagnosis on vehicles
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0102: Carry out service, repair and overhauling of mechanical aggregates in vehicles

Description

This OS unit is about preparing and executing the service, repair, and overhauling of mechanical aggregates. It includes initial preparations, performing the necessary service and repairs, and completing post-service activities to ensure the aggregates function properly and efficiently.

Scope

The scope covers the following :

- Prepare for service, repair and overhauling of mechanical aggregates
- Perform service, repair and overhauling of mechanical aggregates
- Perform post-service and repair activities

Elements and Performance Criteria

Prepare for service, repair and overhauling of mechanical aggregates

To be competent, the user/individual on the job must be able to:

PC1. Review the job card, and obtain sufficient information about the service or repair to be conducted

PC2. Organise for relevant workshop tools, measuring devices and equipment required for service or repair

PC3. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment

PC4. Adhere to the safety aspects while working on a vehicle Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard etc

PC5. Prepare the vehicle for service and repair

Perform service, repair and overhauling of mechanical aggregates

To be competent, the user/individual on the job must be able to:

PC6. Take precautions to avoid damage to the vehicle and its components while working on repair and services

PC7. Use appropriate workshop tools, measuring devices, equipment, and consumables required as per Original Equipment Manufacturer (OEM) Standard Operating Procedure (SOP)

PC8. Remove parts relevant to various mechanical aggregates of the vehicle and place them securely as specified by OEM Mechanical system or aggregates: e.g., drive line, running systems, etc. including power-assisted braking & steering systems etc

PC9. Dismantle the entire mechanical aggregate, if required, and report the additional repair requirement to the concerned person

PC10. Perform repair and replacement of mechanical system or aggregates

PC11. Perform calibration, overhauling of mechanical system or aggregates

PC12. Clean and condition dismantled components including mechanical aggregates and other accessories before assembly

Perform post-service and repair activities

To be competent, the user/individual on the job must be able to:

PC13. Check the performance of vehicle/aggregate post repair to confirm fault resolution

PC14. Ensure completeness of repair and service task assigned before releasing the vehicle for the next procedure

PC15. Dispose of materials such as waste oil etc. as per the organization's policies

PC16. Keep the failed parts/aggregates, as per organisational policies to show customers the aggregates that have been replaced

PC17. Update the job card, wherever required, to inform the customer about the condition of the vehicle or parts which are relevant to the customer's knowledge

PC18. Ensure all tools, auxiliary material and other equipment are replaced in their respective storage areas before taking up another task

PC19. Return leftover consumables and parts and report if any defects were observed in them to the person concerned

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Standard Operating Procedures (SOP) and the checklists recommended by the Original Equipment Manufacturer (OEM) /auto component manufacturer/dealer for assessing service and repair

KU2. Standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of mechanical components/aggregate in the vehicle

KU3. Safety aspects recommended by OEM and dealer while working on service and repair of electric vehicle

KU4. Electrical hazards, protective measures and first aid: in case of electric shock, electrical arc in public grid or two/three wheeler electric vehicle, the impact of electric current/arc, secondary accidents

KU5. SOP recommended by OEM for using tools/equipment service and repair such as special service tools, measuring instruments, etc

KU6. Different components and aggregates of 2 or 3 wheeler electric vehicles as well as auto component manufacturer's specifications for the same

KU7. Basic technology used in various mechanical systems and components of the 2 and 3 wheeler electric vehicle system and their functioning

KU8. Systems and Components: e.g. power train, running system, various lubrication systems, hydraulic/pneumatic systems, cooling system, etc

KU9. Interconnection of systems with each other and the effect of one system on other systems

KU10. Various sources of information available for assessing servicing and repair of vehicle

KU11. Various servicing procedures for mechanical systems or aggregates

KU12. Common repairs performed on mechanical systems or aggregates of 2 and 3 wheeler electric vehicles

KU13. Various methods for removal, dismantling, cleaning, adjusting, reassembling and testing of mechanical components for proper functioning

KU14. Type and quality of consumables/materials used for the job such as seals, sealant, fasteners, lubricants etc

KU15. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation

KU16. Safely keeping the materials that have been replaced for customer's knowledge

KU17. Importance of keeping notes on the job card and sharing information with the customer about the condition of the vehicle or components

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare for service, repair and overhauling of mechanical aggregates	**10**	**20**	**-**	**-**
PC1. Review the job card, and obtain sufficient information about the service or repair to be conducted	-	-	-	-
PC2. Organise for relevant workshop tools, measuring devices and equipment required for service or repair	-	-	-	-
PC3. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment	-	-	-	-
PC4. Adhere to the safety aspects while working on a vehicle Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard etc	-	-	-	-
PC5. Prepare the vehicle for service and repair	-	-	-	-
Perform service, repair and overhauling of mechanical aggregates	**10**	**25**	**-**	**-**
PC6. Take precautions to avoid damage to the vehicle and its components while working on repair and services	-	-	-	-
PC7. Use appropriate workshop tools, measuring devices, equipment, and consumables required as per Original Equipment Manufacturer (OEM) Standard Operating Procedure (SOP)	-	-	-	-
PC8. Remove parts relevant to various mechanical aggregates of the vehicle and place them securely as specified by OEM Mechanical system or aggregates: e.g., drive line, running systems, etc. including power-assisted braking & steering systems etc	-	-	-	-
PC9. Dismantle the entire mechanical aggregate, if required, and report the additional repair requirement to the concerned person	-	-	-	-
PC10. Perform repair and replacement of mechanical system or aggregates	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Perform calibration, overhauling of mechanical system or aggregates	-	-	-	-
PC12. Clean and condition dismantled components including mechanical aggregates and other accessories before assembly	-	-	-	-
Perform post-service and repair activities	**10**	**25**	**-**	**-**
PC13. Check the performance of vehicle/aggregate post repair to confirm fault resolution	-	-	-	-
PC14. Ensure completeness of repair and service task assigned before releasing the vehicle for the next procedure	-	-	-	-
PC15. Dispose of materials such as waste oil etc. as per the organization's policies	-	-	-	-
PC16. Keep the failed parts/aggregates, as per organisational policies to show customers the aggregates that have been replaced	-	-	-	-
PC17. Update the job card, wherever required, to inform the customer about the condition of the vehicle or parts which are relevant to the customer’s knowledge	-	-	-	-
PC18. Ensure all tools, auxiliary material and other equipment are replaced in their respective storage areas before taking up another task	-	-	-	-
PC19. Return leftover consumables and parts and report if any defects were observed in them to the person concerned	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0102
NOS Name	Carry out service, repair and overhauling of mechanical aggregates in vehicles
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0103: Carry out service, repair and overhauling of electrical and electronic aggregates in vehicles

Description

This OS unit is about performing, and completing service, repair, and overhauling of electrical and electronic aggregates. It covers setup, execution of tasks, and post-service activities to ensure proper functionality and efficiency.

Scope

The scope covers the following :

- Prepare for service, repair and overhauling of electrical and electronic aggregates
- Perform service, repair and overhauling of electrical and electronic aggregates
- Perform post-service and repair activities

Elements and Performance Criteria

Prepare for service, repair and overhauling of electrical and electronic aggregates

To be competent, the user/individual on the job must be able to:

PC1. Review the job card, and obtain sufficient information about the service or repair to be conducted

PC2. Organise for relevant workshop tools, measuring devices and equipment required for service or repair of electrical and electronic aggregates

PC3. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment

PC4. Adhere to the safety aspects while working on a vehicle Safety aspects: PPE, High Voltage (HV) precautions, vehicle safeguard etc

PC5. Prepare the vehicle for service and repair

Perform service, repair and overhauling of electrical and electronic aggregates

To be competent, the user/individual on the job must be able to:

PC6. Take precautions to avoid damage to the vehicle and its components while working on repair and services

PC7. Use appropriate workshop tools, measuring devices, equipment, and consumables required for the job activity as per Original Equipment Manufacturer (OEM) Standard Operating Procedure (SOP)

PC8. Perform service and repair-related tasks on the HV system Tasks: general and mechanical tasks on the vehicle which do not require isolation of the HV systems, mechanical work in the non-live state of the HV systems and replacing parts in the live state of the HV system

PC9. Remove parts relevant to various electrical and electronic aggregates of the vehicle, and then place them securely as specified by OEM Electrical and electronic aggregates: e.g. batteries, battery management system, body management system, telematics, air-conditioning systems, active & passive safety system, media and other systems such as electrical machines and devices, generator, DC/AC and DC/DC converters, AC motor, DC motor, charging systems, etc.

PC10. Dismantle the entire electrical and electronic aggregate, if required, and report the additional repair requirement to the concerned person before performing it

PC11. Test electrical and electronic components post removal wherever applicable as per OEM SOP

PC12. Perform repair of direct faults in electrical and electronic aggregates Direct faults: input sensors, output actuators, wiring harnesses, computer systems, calibration/adjustment specifications, component specifications, component assembly, system modification

PC13. Perform calibration, and overhauling of electrical and electronic aggregates

PC14. Clean and condition dismantled components, including electrical/electronic aggregates and other accessories before assembly

Perform post-service and repair activities

To be competent, the user/individual on the job must be able to:

PC15. Check the performance of vehicle/aggregate post repair to confirm fault resolution

PC16. Ensure completeness of repair and service task assigned before releasing the vehicle for the next procedure

PC17. Dispose of materials such as waste oil etc. as per the organization's policies

PC18. Keep the failed parts/aggregates, as per the organisation’s policies to show customers about the aggregates that have been replaced

PC19. Update the job card, wherever required, to inform the customer about the condition of the vehicle or parts which are relevant to the customer’s knowledge

PC20. Ensure all tools, auxiliary material and other equipment are replaced in their respective storage areas before taking up other tasks

PC21. Return leftover consumables and parts and report if any defects were observed in them to the person concerned

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Various sources of information available for assessing servicing and repair of vehicle

KU2. Standard Operating Procedure (SOP) and the checklists recommended by the Original Equipment Manufacturer (OEM) /auto component manufacturer/dealer for assessing service and repair

KU3. Standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of electrical/electronic components/aggregate in the vehicle

KU4. SOP recommended by OEM for using tools/equipment service and repair such as special service tools, measuring instruments, volt meters, ammeters, ohmmeters, battery testers, dedicated and computer-based diagnostic equipment, oscilloscopes etc

KU5. Safety aspects recommended by OEM and dealer while working on service and repair of vehicle

KU6. Electrical hazards, protective measures and first aid: in case of electric shock, electrical arc in public grid or two/three wheeler electric vehicle, the impact of electric current/arc, secondary accidents

KU7. Procedure to work on the High Voltage (HV) systems that do not require isolation, troubleshooting and replacing parts on the active HV system

KU8. Safety rules for electrical work on HV systems before starting the work i.e., isolate, safeguard reconnection, verify the non-live state, earth or short-circuit and shroud or safeguard adjacent live parts

KU9. Different components and aggregates of 2 or 3 wheeler electric vehicles as well as auto component manufacturer's specifications for the same

KU10. Basic technology used in various electrical/electronic systems and components of the vehicle system and their functioning

KU11. Systems and Components: e.g., batteries, battery management system, body management system, telematics, air-conditioning systems, active & passive safety system, media and other systems such as electrical machines and devices, generator, DC/AC and DC/DC converters, AC motor, DC motor, charging systems, etc

KU12. Interconnection of systems with each other and the effect of one system on other systems

KU13. Various methods for removal, dismantling, cleaning, adjusting, reassembling and testing of electrical and electronic components for proper functioning

KU14. Various servicing procedures for electrical and electronic systems or aggregates

KU15. Common repairs performed on electrical and electronic systems or aggregates of 2 and 3 wheeler electric vehicles

KU16. Type and quality of consumables/materials used for the job such as seals, sealant, fasteners, lubricants etc

KU17. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation

KU18. Safely keeping the materials that have been replaced for customer's knowledge

KU19. Importance of putting notes on the job card and sharing information with the customer about the condition of the vehicle or components

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare for service, repair and overhauling of electrical and electronic aggregates	**10**	**20**	**-**	**-**
PC1. Review the job card, and obtain sufficient information about the service or repair to be conducted	-	-	-	-
PC2. Organise for relevant workshop tools, measuring devices and equipment required for service or repair of electrical and electronic aggregates	-	-	-	-
PC3. Ensure proper condition, functioning and calibration of workshop tools, measuring devices and equipment	-	-	-	-
PC4. Adhere to the safety aspects while working on a vehicle Safety aspects: PPE, High Voltage (HV) precautions, vehicle safeguard etc	-	-	-	-
PC5. Prepare the vehicle for service and repair	-	-	-	-
Perform service, repair and overhauling of electrical and electronic aggregates	**10**	**25**	**-**	**-**
PC6. Take precautions to avoid damage to the vehicle and its components while working on repair and services	-	-	-	-
PC7. Use appropriate workshop tools, measuring devices, equipment, and consumables required for the job activity as per Original Equipment Manufacturer (OEM) Standard Operating Procedure (SOP)	-	-	-	-
PC8. Perform service and repair-related tasks on the HV system Tasks: general and mechanical tasks on the vehicle which do not require isolation of the HV systems, mechanical work in the non-live state of the HV systems and replacing parts in the live state of the HV system	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Remove parts relevant to various electrical and electronic aggregates of the vehicle, and then place them securely as specified by OEM Electrical and electronic aggregates: e.g. batteries, battery management system, body management system, telematics, air-conditioning systems, active & passive safety system, media and other systems such as electrical machines and devices, generator, DC/AC and DC/DC converters, AC motor, DC motor, charging systems, etc.	-	-	-	-
PC10. Dismantle the entire electrical and electronic aggregate, if required, and report the additional repair requirement to the concerned person before performing it	-	-	-	-
PC11. Test electrical and electronic components post removal wherever applicable as per OEM SOP	-	-	-	-
PC12. Perform repair of direct faults in electrical and electronic aggregates Direct faults: input sensors, output actuators, wiring harnesses, computer systems, calibration/adjustment specifications, component specifications, component assembly, system modification	-	-	-	-
PC13. Perform calibration, and overhauling of electrical and electronic aggregates	-	-	-	-
PC14. Clean and condition dismantled components, including electrical/electronic aggregates and other accessories before assembly	-	-	-	-
Perform post-service and repair activities	**10**	**25**	**-**	**-**
PC15. Check the performance of vehicle/aggregate post repair to confirm fault resolution	-	-	-	-
PC16. Ensure completeness of repair and service task assigned before releasing the vehicle for the next procedure	-	-	-	-
PC17. Dispose of materials such as waste oil etc. as per the organization's policies	-	-	-	-
PC18. Keep the failed parts/aggregates, as per the organisation’s policies to show customers about the aggregates that have been replaced	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Update the job card, wherever required, to inform the customer about the condition of the vehicle or parts which are relevant to the customer’s knowledge	-	-	-	-
PC20. Ensure all tools, auxiliary material and other equipment are replaced in their respective storage areas before taking up other tasks	-	-	-	-
PC21. Return leftover consumables and parts and report if any defects were observed in them to the person concerned	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0103
NOS Name	Carry out service, repair and overhauling of electrical and electronic aggregates in vehicles
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0104: Carry out battery swapping and management for 2 and 3 wheeler Electric vehicle

Description

This OS unit is about evaluating the performance of battery storage and management systems, operating and troubleshooting the EV charging ecosystem, managing EV charging stations, and overseeing battery swapping processes.

Scope

The scope covers the following :

- Check the performance of the battery storage system and battery management system
- Operate and troubleshoot the EV charging ecosystem
- Manage EV charging station
- Manage battery swapping

Elements and Performance Criteria

Check the performance of the battery storage system and battery management system

To be competent, the user/individual on the job must be able to:

PC1. Ensure the safety aspects are taken care of before checking the performance of the battery storage system and battery management system

PC2. Perform verification of cell performance against supplier datasheet

PC3. Replace a damaged cell with the same resistance cell

PC4. Perform Battery Management System (BMS) interfacing with Battery Pack configuration on software application

PC5. Ensure that all connections are made with BMS to each cell of the battery

PC6. Carry out voltage, current and temperature measurements with BMS

PC7. Verify mapping for charging and discharging

PC8. Handle the storage of voltage battery systems

PC9. Dispose of high-voltage battery systems

PC10. Test and repair high-voltage battery systems

PC11. Perform replacement of defective Battery Module of 48V Module Systems

PC12. Perform the diagnosis and testing of EV battery controls

PC13. Perform the repair of EV battery controls

Operate and troubleshoot the EV charging ecosystem

To be competent, the user/individual on the job must be able to:

PC14. Identify the type of charger and voltage levels that operate standard chargers

PC15. Determine the charging time and charging inputs under various conditions

PC16. Perform the diagnosis and take remedial action for the charger not responding and not delivering the expected current

PC17. Plan work in compliance with standard safety norms if not fix the replace the charger

Manage EV charging station

To be competent, the user/individual on the job must be able to:

PC18. Ensure the safety aspects are taken care of before accessing the EV charging station

PC19. Ensure that the EV charging station is in working condition

PC20. Ensure to replace the equipment and other consumables for accessibility of the EV charging station

PC21. Manage the token system at the EV charging station

PC22. Explain the compatibility, plans, charging time etc. to the customer who wants to access the EV charging station for charging

PC23. Ensure that customer queries have been answered

PC24. Ensure that the voltage, other specifications and other regulations prescribed by the Original Equipment Manufacturer (OEM) are matching with the EV charging station specification before charging

PC25. Take confirmation from the customer before using the EV charging station for charging the EV

PC26. Ensure the vehicle is parked in the safe area once the EV is fully charged

PC27. Inform the customer once the charging is completed

PC28. Manage the billing counter of the EV charging station

Manage battery swapping

To be competent, the user/individual on the job must be able to:

PC29. Ensure the stocks of fully charged batteries for swapping

PC30. Ensure the batteries are kept in a safe place

PC31. Explain the plans, pricing and compatibility of the batteries for swapping

PC32. Ensure that the voltage, other specifications and other regulations prescribed by the Original Equipment Manufacturer (OEM) of the battery are matching with the battery to be swapped

PC33. Ensure that customer queries have been answered

PC34. Take confirmation from the customer before swapping

PC35. Ensure the battery is swapped safely

PC36. Manage the billing counter of the battery swap

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The safety aspects related to battery management, battery charging, EV charging and battery stations

KU2. The token systems at EV charging and swapping stations

KU3. Understanding of different cell chemistries geometries and various sensors installed in battery management systems

KU4. Battery Management System (BMS) with Battery Pack configuration on software application

KU5. Measurements of voltage, current and temperature with BMS

KU6. Dispose of techniques of high voltage battery systems techniques

KU7. Type of charger and voltage levels operate standard chargers

KU8. Diagnosis and take remedial action for charger not responding, charger not delivering expected current

KU9. Specifications of the EV charging stations

KU10. Compatibility of EV for charging in EV charging stations

KU11. Various plans, charging time etc. for an EV at an EV charging station

KU12. Billing management for EV charging and swapping stations

KU13. Stocking methods of batteries for swapping

KU14. Compatibility of battery for battery swapping in vehicle

KU15. Various plans and pricing of battery swapping

KU16. Probing techniques to understand the customer's requirements

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Check the performance of the battery storage system and battery management system	**5**	**10**	**-**	**-**
PC1. Ensure the safety aspects are taken care of before checking the performance of the battery storage system and battery management system	-	-	-	-
PC2. Perform verification of cell performance against supplier datasheet	-	-	-	-
PC3. Replace a damaged cell with the same resistance cell	-	-	-	-
PC4. Perform Battery Management System (BMS) interfacing with Battery Pack configuration on software application	-	-	-	-
PC5. Ensure that all connections are made with BMS to each cell of the battery	-	-	-	-
PC6. Carry out voltage, current and temperature measurements with BMS	-	-	-	-
PC7. Verify mapping for charging and discharging	-	-	-	-
PC8. Handle the storage of voltage battery systems	-	-	-	-
PC9. Dispose of high-voltage battery systems	-	-	-	-
PC10. Test and repair high-voltage battery systems	-	-	-	-
PC11. Perform replacement of defective Battery Module of 48V Module Systems	-	-	-	-
PC12. Perform the diagnosis and testing of EV battery controls	-	-	-	-
PC13. Perform the repair of EV battery controls	-	-	-	-
Operate and troubleshoot the EV charging ecosystem	**5**	**10**	**-**	**-**
PC14. Identify the type of charger and voltage levels that operate standard chargers	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Determine the charging time and charging inputs under various conditions	-	-	-	-
PC16. Perform the diagnosis and take remedial action for the charger not responding and not delivering the expected current	-	-	-	-
PC17. Plan work in compliance with standard safety norms if not fix the replace the charger	-	-	-	-
Manage EV charging station	**10**	**25**	**-**	**-**
PC18. Ensure the safety aspects are taken care of before accessing the EV charging station	-	-	-	-
PC19. Ensure that the EV charging station is in working condition	-	-	-	-
PC20. Ensure to replace the equipment and other consumables for accessibility of the EV charging station	-	-	-	-
PC21. Manage the token system at the EV charging station	-	-	-	-
PC22. Explain the compatibility, plans, charging time etc. to the customer who wants to access the EV charging station for charging	-	-	-	-
PC23. Ensure that customer queries have been answered	-	-	-	-
PC24. Ensure that the voltage, other specifications and other regulations prescribed by the Original Equipment Manufacturer (OEM) are matching with the EV charging station specification before charging	-	-	-	-
PC25. Take confirmation from the customer before using the EV charging station for charging the EV	-	-	-	-
PC26. Ensure the vehicle is parked in the safe area once the EV is fully charged	-	-	-	-
PC27. Inform the customer once the charging is completed	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Manage the billing counter of the EV charging station	-	-	-	-
Manage battery swapping	**10**	**25**	**-**	**-**
PC29. Ensure the stocks of fully charged batteries for swapping	-	-	-	-
PC30. Ensure the batteries are kept in a safe place	-	-	-	-
PC31. Explain the plans, pricing and compatibility of the batteries for swapping	-	-	-	-
PC32. Ensure that the voltage, other specifications and other regulations prescribed by the Original Equipment Manufacturer (OEM) of the battery are matching with the battery to be swapped	-	-	-	-
PC33. Ensure that customer queries have been answered	-	-	-	-
PC34. Take confirmation from the customer before swapping	-	-	-	-
PC35. Ensure the battery is swapped safely	-	-	-	-
PC36. Manage the billing counter of the battery swap	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0104
NOS Name	Carry out battery swapping and management for 2 and 3 wheeler Electric vehicle
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0105: Manage vehicle service and repair workshops

Description

This OS unit is about managing the workshop area by preparing a comprehensive workshop plan, overseeing workshop processes, and efficiently handling tools, equipment, and spare parts.

Scope

The scope covers the following :

- Manage workshop area
- Prepare a workshop plan
- Manage workshop processes
- Manage tools, equipment and spare parts

Elements and Performance Criteria

Manage workshop area

To be competent, the user/individual on the job must be able to:

PC1. Prepare bay space management plan to ensure optimised placement of vehicles, quick service/repair, ease of access, availability of tools and equipment, adequate ventilation, adequate charging stations, safety, security, etc

PC2. Inspect the bay area to ensure adherence to the plan

PC3. Ensure the work area is kept clean and tidy

Prepare a workshop plan

To be competent, the user/individual on the job must be able to:

PC4. Check the various types of requests daily Type of request: e.g., service, repair and complaint etc

PC5. Plan work on various types of requests according to the schedule and location of the requests.Type of requests: e.g., service, repair and complaint etc.Location of the requests: e.g., at the customer site, on road, in the workshop etc.

PC6. Prioritize the various types of requests with respect to technician, spare parts availability, schedule and location etc. Type of requests: e.g., service, repair and complaint etc.Location of the requests: e.g., at the customer site, on road, in the workshop, etc

PC7. Perform workshop load calculation to plan daily operations and their execution in a smooth manner

Manage workshop processes

To be competent, the user/individual on the job must be able to:

PC8. Implement and maintain safety systems and processes Safety aspects: Availability and use of PPE, HV (High Voltage) safety measures, vehicle safeguards, hazard review and management, etc

PC9. Optimise processes at the various bays (including AMC bay, quick repair bay etc.) of the workshop to ensure smooth operations and adherence to quality standards

PC10. Ensure the availability of standard operating procedures for all key processes

PC11. Check adherence to processes

PC12. Ensure all deviations are duly approved and documented

PC13. Prioritize the service or repair requirements and allocate resources based on various parameters .Parameters: workload, resources available, constraints

PC14. Supervise the team in resolving critical issues that need urgent action or specialist intervention

PC15. Perform the final inspection after the service and repair operations are completed, in case the quality inspector is unavailable

PC16. Manage quality issues to reduce rework or repeat complaints

PC17. Prepare all diagnostic and repair reports in coordination with components specialists and technical manager for future reference

PC18. Report to the service manager or GM Service on the functioning of the workshop or body shop and any requirements, challenges and problems faced

Manage tools, equipment and spare parts

To be competent, the user/individual on the job must be able to:

PC19. Manage the maintenance of workshop facilities and other tools including fixed equipment

PC20. Ensure the supply of materials, parts and other requirements in coordination with the spare parts manager

PC21. Ensure procurement of urgently required spare parts from the market in the most cost-effective manner which is not available at the workshop spare counter

PC22. Get the replacement of failed parts/aggregates in coordination with the warranty processor as per the warranty manual and organisational guidelines

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Structure of a typical 2 and 3 wheeler Electric vehicle service and repair workshop

KU2. Various workshop departments and the interlinkage of these departments in service, repair, and fault diagnosis

KU3. Bay space management plan and ways to optimise bay space

KU4. Various types of service and repair requests and complaints

KU5. Importance of prioritizing work and key factors that decide the priority

KU6. Workshop load calculation

KU7. Various operations in a service and repair workshop

KU8. Planning, and prioritising tasks of the team in the workshop

KU9. Safety regulation in automotive workshop

KU10. Safety systems and processes in a service and repair workshop

KU11. Strategies to optimise processes at the various bays

KU12. Importance of standard operating procedures

KU13. Importance and process of inspection after the service and repair operations

KU14. Common quality issues that need to be managed

KU15. Elements in a diagnostic and repair report

KU16. Importance of reporting any requirements, challenges and problems faced by management

KU17. Tools, equipment, spare parts and materials needed in a 2 and 3 wheeler service and repair workshop

KU18. Maintenance requirements in various tools and equipment

KU19. Procurement process and considerations

KU20. Negotiation with vendors

KU21. Warranty and insurance policies and related procedures

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Manage workshop area	**5**	**10**	**-**	**-**
PC1. Prepare bay space management plan to ensure optimised placement of vehicles, quick service/repair, ease of access, availability of tools and equipment, adequate ventilation, adequate charging stations, safety, security, etc	-	-	-	-
PC2. Inspect the bay area to ensure adherence to the plan	-	-	-	-
PC3. Ensure the work area is kept clean and tidy	-	-	-	-
Prepare a workshop plan	**5**	**10**	**-**	**-**
PC4. Check the various types of requests daily Type of request: e.g., service, repair and complaint etc	-	-	-	-
PC5. Plan work on various types of requests according to the schedule and location of the requests.Type of requests: e.g., service, repair and complaint etc.Location of the requests: e.g., at the customer site, on road, in the workshop etc.	-	-	-	-
PC6. Prioritize the various types of requests with respect to technician, spare parts availability, schedule and location etc. Type of requests: e.g., service, repair and complaint etc.Location of the requests: e.g., at the customer site, on road, in the workshop, etc	-	-	-	-
PC7. Perform workshop load calculation to plan daily operations and their execution in a smooth manner	-	-	-	-
Manage workshop processes	**10**	**25**	**-**	**-**
PC8. Implement and maintain safety systems and processes Safety aspects: Availability and use of PPE, HV (High Voltage) safety measures, vehicle safeguards, hazard review and management, etc	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Optimise processes at the various bays (including AMC bay, quick repair bay etc.) of the workshop to ensure smooth operations and adherence to quality standards	-	-	-	-
PC10. Ensure the availability of standard operating procedures for all key processes	-	-	-	-
PC11. Check adherence to processes	-	-	-	-
PC12. Ensure all deviations are duly approved and documented	-	-	-	-
PC13. Prioritize the service or repair requirements and allocate resources based on various parameters .Parameters: workload, resources available, constraints	-	-	-	-
PC14. Supervise the team in resolving critical issues that need urgent action or specialist intervention	-	-	-	-
PC15. Perform the final inspection after the service and repair operations are completed, in case the quality inspector is unavailable	-	-	-	-
PC16. Manage quality issues to reduce rework or repeat complaints	-	-	-	-
PC17. Prepare all diagnostic and repair reports in coordination with components specialists and technical manager for future reference	-	-	-	-
PC18. Report to the service manager or GM Service on the functioning of the workshop or body shop and any requirements, challenges and problems faced	-	-	-	-
Manage tools, equipment and spare parts	**10**	**25**	**-**	**-**
PC19. Manage the maintenance of workshop facilities and other tools including fixed equipment	-	-	-	-
PC20. Ensure the supply of materials, parts and other requirements in coordination with the spare parts manager	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Ensure procurement of urgently required spare parts from the market in the most cost-effective manner which is not available at the workshop spare counter	-	-	-	-
PC22. Get the replacement of failed parts/aggregates in coordination with the warranty processor as per the warranty manual and organisational guidelines	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0105
NOS Name	Manage vehicle service and repair workshops
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0107: Manage the vehicle service and repair teams

Description

This OS unit is about allocating work, monitoring progress, evaluating performance, and handling escalations.

Scope

The scope covers the following :

- Allocate work
- Monitor work
- Evaluate performance
- Deal with escalation

Elements and Performance Criteria

Allocate work

To be competent, the user/individual on the job must be able to:

PC1. Allocate work to the team based on priorities, workload, resource availability etc. Resources: e.g., tools, equipment, materials, time, budgets, support staff, etc.

PC2. Create a time-table and work allocation plan

PC3. Communicate work requirements clearly to the team members verbally and by sharing the work allocation plan

Monitor team

To be competent, the user/individual on the job must be able to:

PC4. Review the job cards and ensure the service and repair staff has filled in the information correctly

PC5. Review and validate the final proposal and quotation created by the service

PC6. Validate the options for repair or replacement made by the service technician or lead service technician

PC7. Ensure that team members use the equipment and special tools correctly as per the OEM guidelines

PC8. Ensure the team is safe while working on the vehicle. Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard etc

PC9. Ensure that team members take precautions to avoid damage to the vehicle and its components while working

PC10. Ensure that technicians follow the correct procedures as per SOP

PC11. Provide the solutions to work-related problems and concerns raised by team members

PC12. Ensure discipline and adherence to workshop rules within the team

PC13. Take attendance and approve leaves in line with HR policies

PC14. Ensure that the team uses resources in a responsible manner

PC15. Validate the post-repair activities of the team Post-repair activities: e.g. test drive the vehicle to ensure correct work is done no work is left on a vehicle, ensure the team dispose off the required materials after the work, ensure the team places the tools/equipment in an organised manner to maintain safe and tidy workstation, ensure the team keep the failed parts/aggregates, ensure the team puts a comment on the job card to inform the customer about condition of the vehicle etc.

PC16. Explain the relevant information of the service/repair to the customer

Evaluate performance

To be competent, the user/individual on the job must be able to:

PC17. Document all performance indicators and metrics of subordinates in the prescribed format of the organisation

PC18. Set goals and targets as per organisational directives for the team members

PC19. Inform the team about appraisal-related process flow as per respective performance documents

PC20. Evaluate the performance of subordinates and reporting executives on the designed measures and metrics as per the guidelines of the organisation

PC21. Document all Key Performance Indicators (KPIs) and metrics of the team in the prescribed format of the organization

PC22. Share performance evaluation results along with feedback on how they can do better

PC23. Take feedback from the team on how to improve work performance

PC24. Agree on an action plan for improvement

PC25. Handover all the documents and appropriate support measures to the human resources department for official records

Deal with escalation

To be competent, the user/individual on the job must be able to:

PC26. Get the escalation document filled as per your organisational guidelines

PC27. Prioritise the escalation with the current line of tasks

PC28. Analyse the root cause of the escalation referring to various parameters Parameters: e.g., the vehicle was serviced/repaired correctly in the first go, this is a new problem, the old problem was not fixed correctly etc

PC29. Provide a solution based on the root cause identified, the additional cost (if any), and the time required to resolve the issue with justification as per your organisation’s policy

PC30. Apply conflict resolution techniques in situations where the need arises

PC31. Take the confirmation from the stakeholder on the solution provided

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Various tools and equipment used in the automotive workshop

KU2. Maintenance requirement in various tools and equipment

KU3. Safety and health policies and regulations for the workplace as well as for Automotive trade in general

KU4. Various departments and areas of the workshop and the interlinkage of these aspects in service, repair and fault diagnosis

KU5. Workload calculation, task prioritisation

KU6. Escalations and complaint resolution

KU7. Communication etiquette with the team and other departments

KU8. Fault diagnosis, service and repair procedures on vehicle

KU9. Team handling and conflict resolution

KU10. Precautions to avoid damage to the vehicle and its components while working

KU11. SOP and the checklist recommended by the Original Equipment Manufacturer (OEM) /auto component manufacturer/dealer for assessing service and repair

KU12. Standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of echanical/electrical/electronic components/aggregate in the vehicle

KU13. Various methods for removal, dismantling, cleaning, adjusting, reassembling and testing of mechanical/electrical/electronic components for proper functioning

KU14. Safety aspects recommended by OEM and dealer while working on service and repair of vehicle

KU15. Electrical hazards, protective measures and first aid: in case of electric shock, electrical arc in public grid or two/three wheeler electric vehicle, the impact of electric current/arc, secondary accidents

KU16. Five safety rules for electrical work on HV systems before starting the work i.e. isolate, safeguard reconnection, verify the non-live state, earth or short-circuit and shroud or safeguard adjacent live parts

KU17. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation

KU18. Safely keeping the materials that have been replaced for customer's knowledge

KU19. Importance of keeping putting notes on the job card and sharing information with the customer about the condition of the vehicle or components

KU20. SOPs of the organisation for appraisals, incentives, promotions and performance evaluation

KU21. Reports to be created as part of roles and responsibilities

KU22. Software or format such as MS Office and Management Information System (MIS) as prescribed by the organisation

KU23. Organisation's incentive policy and procedures

KU24. Institutional and professional code of ethics and standards of practice

KU25. Framework and guidelines prescribed by the organisation for performance evaluations and appraisals based on the same

KU26. Process flow for performance evaluation, documentation and related appraisals

KU27. SOPs for query and problem reporting and their redressal in the organisation framework and guidelines prescribed by the organisation for query and problem redressal

KU28. Redressal documentation mechanisms available in the organisation

KU29. Documentation requirements for appraisals and other performance evaluations of various subordinate positions

KU30. Subordinate and report the executives problems and queries and document them in the organisations prescribed format

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Allocate work	**5**	**10**	**-**	**-**
PC1. Allocate work to the team based on priorities, workload, resource availability etc. Resources: e.g., tools, equipment, materials, time, budgets, support staff, etc.	-	-	-	-
PC2. Create a time-table and work allocation plan	-	-	-	-
PC3. Communicate work requirements clearly to the team members verbally and by sharing the work allocation plan	-	-	-	-
Monitor team	**5**	**10**	**-**	**-**
PC4. Review the job cards and ensure the service and repair staff has filled in the information correctly	-	-	-	-
PC5. Review and validate the final proposal and quotation created by the service	-	-	-	-
PC6. Validate the options for repair or replacement made by the service technician or lead service technician	-	-	-	-
PC7. Ensure that team members use the equipment and special tools correctly as per the OEM guidelines	-	-	-	-
PC8. Ensure the team is safe while working on the vehicle. Safety aspects: PPE, HV (High Voltage) precautions, vehicle safeguard etc	-	-	-	-
PC9. Ensure that team members take precautions to avoid damage to the vehicle and its components while working	-	-	-	-
PC10. Ensure that technicians follow the correct procedures as per SOP	-	-	-	-
PC11. Provide the solutions to work-related problems and concerns raised by team members	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Ensure discipline and adherence to workshop rules within the team	-	-	-	-
PC13. Take attendance and approve leaves in line with HR policies	-	-	-	-
PC14. Ensure that the team uses resources in a responsible manner	-	-	-	-
PC15. Validate the post-repair activities of the team Post-repair activities: e.g. test drive the vehicle to ensure correct work is done no work is left on a vehicle, ensure the team dispose off the required materials after the work, ensure the team places the tools/equipment in an organised manner to maintain safe and tidy workstation, ensure the team keep the failed parts/aggregates, ensure the team puts a comment on the job card to inform the customer about condition of the vehicle etc.	-	-	-	-
PC16. Explain the relevant information of the service/repair to the customer	-	-	-	-
Evaluate performance	**10**	**25**	**-**	**-**
PC17. Document all performance indicators and metrics of subordinates in the prescribed format of the organisation	-	-	-	-
PC18. Set goals and targets as per organisational directives for the team members	-	-	-	-
PC19. Inform the team about appraisal-related process flow as per respective performance documents	-	-	-	-
PC20. Evaluate the performance of subordinates and reporting executives on the designed measures and metrics as per the guidelines of the organisation	-	-	-	-
PC21. Document all Key Performance Indicators (KPIs) and metrics of the team in the prescribed format of the organization	-	-	-	-
PC22. Share performance evaluation results along with feedback on how they can do better	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Take feedback from the team on how to improve work performance	-	-	-	-
PC24. Agree on an action plan for improvement	-	-	-	-
PC25. Handover all the documents and appropriate support measures to the human resources department for official records	-	-	-	-
Deal with escalation	**10**	**25**	**-**	**-**
PC26. Get the escalation document filled as per your organisational guidelines	-	-	-	-
PC27. Prioritise the escalation with the current line of tasks	-	-	-	-
PC28. Analyse the root cause of the escalation referring to various parameters Parameters: e.g., the vehicle was serviced/repaired correctly in the first go, this is a new problem, the old problem was not fixed correctly etc	-	-	-	-
PC29. Provide a solution based on the root cause identified, the additional cost (if any), and the time required to resolve the issue with justification as per your organisation’s policy	-	-	-	-
PC30. Apply conflict resolution techniques in situations where the need arises	-	-	-	-
PC31. Take the confirmation from the stakeholder on the solution provided	-	-	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0107
NOS Name	Manage the vehicle service and repair teams
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

MSU/ASC/N0106: Deal with Customer's Request and Complaints

Description

This OS unit is about clarifying and addressing customer requests or complaints effectively. It involves managing these issues promptly and taking proactive steps to refine processes, aiming to reduce the occurrence of future complaints and improve overall customer satisfaction.

Scope

The scope covers the following :

- Obtain clarity about the customer request or complaint
- Deal with customer requests and complaints
- Take measures to improve processes to reduce customer complaints

Elements and Performance Criteria

Obtain clarity about the customer request or complaint

To be competent, the user/individual on the job must be able to:

PC1. Connect with the customer face-to-face or telephonically

PC2. Identify the type of special request or complaint of the customer

PC3. Seek detailed information about what has happened and what is expected by the customer from the organisation

PC4. Cross-check your understanding by para-phrasing to the customer

PC5. Document the customer request and complaint as per organisational guidelines

Deal with customer requests and complaints

To be competent, the user/individual on the job must be able to:

PC6. Assess the requests or complaints by verifying facts

PC7. Prioritise the escalation with the current line of tasks

PC8. Analyse the root cause of complaints referring to work performed or work to be performed

PC9. Identify all the possible solutions and the best possible solution that can be offered to the customer within organisational policies

PC10. Validate the possible solutions and the cost associated by coordinating with multiple departments Coordinate with Departments: e.g. Customer service, service and repair, insurance, spare parts, billing etc

PC11. Discuss the best possible solution for the requests and the cost associated with the customer

PC12. Take customer agreement on the solution offered

PC13. Allocate the resource to service the request or resolve the problem as per the solution agreed upon

PC14. Monitor the progress of the solution being implemented and ensure adherence to agreed timelines

PC15. Inform customer of the resolution and take their feedback

Take measures to improve processes to reduce customer complaints

To be competent, the user/individual on the job must be able to:

PC16. Document all customer special requests and complaints and solutions offered, and their feedback

PC17. Analyse types of problems that recur

PC18. Identify measures that can reduce the occurrence of the problem

PC19. Seek approval from management to implement measures identified that can reduce the occurrence of the problem

PC20. Implement approved measures in coordination with the team and make respective updates in SOPs, policies, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Etiquettes related to telephonic conversation and face-to-face discussion

KU2. Understanding of various departments

KU3. Common types of customer requests and complaints at the workplace

KU4. Importance of taking complete information regarding customer requests or complaints, cross-checking its understanding and documenting the information obtained

KU5. Sources of information to verify facts provided by the customer

KU6. Need for prioritizing escalations

KU7. Compliant handling process

KU8. Concepts and tools and parameters for root cause analysis

KU9. Problem-solving strategies

KU10. Cost estimation for solutions to common problems

KU11. Role of various departments in providing a complete solution to the customer

KU12. Importance of taking customer agreement on the solution offered

KU13. Importance of allocating the right resource and monitoring the progress of the solution

KU14. Importance and process to ensure measures are taken to improve processes to reduce customer complaints

KU15. Purpose of documentation: e.g -For skill gap analysis of the technician associated (if any) , a new type of problem (if observed for the first time) , OEM has a defective batch while manufacturing etc

KU16. Cross-selling techniques

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Obtain clarity about the customer request or complaint	**10**	**10**	**-**	**-**
PC1. Connect with the customer face-to-face or telephonically	-	-	-	-
PC2. Identify the type of special request or complaint of the customer	-	-	-	-
PC3. Seek detailed information about what has happened and what is expected by the customer from the organisation	-	-	-	-
PC4. Cross-check your understanding by para-phrasing to the customer	-	-	-	-
PC5. Document the customer request and complaint as per organisational guidelines	-	-	-	-
Deal with customer requests and complaints	**20**	**20**	**-**	**-**
PC6. Assess the requests or complaints by verifying facts	-	-	-	-
PC7. Prioritise the escalation with the current line of tasks	-	-	-	-
PC8. Analyse the root cause of complaints referring to work performed or work to be performed	-	-	-	-
PC9. Identify all the possible solutions and the best possible solution that can be offered to the customer within organisational policies	-	-	-	-
PC10. Validate the possible solutions and the cost associated by coordinating with multiple departments Coordinate with Departments: e.g. Customer service, service and repair, insurance, spare parts, billing etc	-	-	-	-
PC11. Discuss the best possible solution for the requests and the cost associated with the customer	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Take customer agreement on the solution offered	-	-	-	-
PC13. Allocate the resource to service the request or resolve the problem as per the solution agreed upon	-	-	-	-
PC14. Monitor the progress of the solution being implemented and ensure adherence to agreed timelines	-	-	-	-
PC15. Inform customer of the resolution and take their feedback	-	-	-	-
Take measures to improve processes to reduce customer complaints	**20**	**20**	**-**	**-**
PC16. Document all customer special requests and complaints and solutions offered, and their feedback	-	-	-	-
PC17. Analyse types of problems that recur	-	-	-	-
PC18. Identify measures that can reduce the occurrence of the problem	-	-	-	-
PC19. Seek approval from management to implement measures identified that can reduce the occurrence of the problem	-	-	-	-
PC20. Implement approved measures in coordination with the team and make respective updates in SOPs, policies, etc.	-	-	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ASC/N0106
NOS Name	Deal with Customer's Request and Complaints
Sector	Automotive
Sub-Sector	
Occupation	Automotive Servicing and Repair
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	30/08/2026
NSQC Clearance Date	31/08/2023

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

• Batches assigned to the assessment agencies for conducting the assessment on SIP or email

• Assessment agencies send the assessment confirmation to VTP/TC looping Madhavi Skills University (MSU)

• Assessment agency deploys the ToA-certified Assessor for executing the assessment

• MSU monitors the assessment process & records

2. Testing Environment:

• Check the Assessment location, date and time

• If the batch size is more than 30, then there should be 2 Assessors.

• Check that the allotted time for the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

• Question bank is created by the Subject Matter Experts (SME) are verified by the other SMEs

• Questions are mapped to the specified assessment criteria

• Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

• Time-stamped & geotagged reporting of the assessor from the assessment location

• Centre photographs with signboards and scheme-specific branding

5. Method of verification or validation:

• Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

• Hard and copies of the documents are stored

On the Job:

1. Each module (which covers the job profile) will be assessed separately.

2. The candidate must score 60% in each module to complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and aptitude are:

• Videos of Trainees during OJT

• Portfolio of items produced

• Viva

• Demonstrations

4. Assessment of each Module will ensure that the candidate is able to:

• Perform the skills mentioned in the assessment criteria.

• Apply underpinning knowledge and understanding to the tasks performed to achieve desired level of competencies

• Demonstrate behavioral and attitudinal soft skills as mentioned in the assessment criteria.

Minimum Aggregate Passing % at QP Level : 60

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSU/ASC/N0101.Perform fault diagnosis on vehicles	30	70	-	-	100	15
MSU/ASC/N0102.Carry out service, repair and overhauling of mechanical aggregates in vehicles	30	70	-	-	100	15
MSU/ASC/N0103.Carry out service, repair and overhauling of electrical and electronic aggregates in vehicles	30	70	-	-	100	20
MSU/ASC/N0104.Carry out battery swapping and management for 2 and 3 wheeler Electric vehicle	30	70	-	-	100	10
MSU/ASC/N0105.Manage vehicle service and repair workshops	30	70	-	-	100	10
MSU/ASC/N0107.Manage the vehicle service and repair teams	30	70	-	-	100	10
MSU/ASC/N0106.Deal with Customer's Request and Complaints	50	50	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**250**	**500**	**-**	**-**	**750**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.